Junho Andrew Choi

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SUMMARY

Data Science and Integrative Biology senior at UC Berkeley targeting data science, analytics, and data engineering roles. Mapped and documented a production ERP schema, audited data quality in a Neo4j graph pipeline, built ingestion pipelines across clinical modalities, and ran survey research that shaped product decisions, with a habit of checking whether a result actually replicates before reporting it.

EDUCATION

University of California, Berkeley Expected May 2027
B.A. Data Science (Concentration: Computational Methods in Biology) and B.A. Integrative Biology *Berkeley, CA*
Coursework: Data Engineering (DS C101, in progress), Machine Learning (CS 189, in progress), Probability for Data Science (Data C140), Principles & Techniques of Data Science (Data 100), Data Structures (CS 61B), Optimization Modeling

SKILLS

Languages: Python (pandas, NumPy, scikit-learn), SQL, Java, R
Data & Statistics: PostgreSQL, Neo4j, SQLite, schema mapping, data modeling, ingestion pipelines, data quality auditing, entity resolution, probability, hypothesis testing, regression, exploratory data analysis (EDA), cohort segmentation, survey design, data visualization
Tools: Git, Jupyter, PyTorch, React, Vite, Figma
Spoken: Korean (native), English (fluent), Mandarin

EXPERIENCE

SmartMind AI May 2026 - Aug 2026
AI Researcher, Data Science Team *Seoul, Republic of Korea*
- Initiated and led a natural-language-to-SQL pipeline on a live enterprise ERP system, mapping the 6 core tables of 20 required for coverage and replacing single-run pass/fail testing with a multi-trial measurement design over a fixed golden set of query scenarios, reporting a measured pass rate with run-to-run variance
- Traced silently failing queries back to the Neo4j ontology (ApeRAG/Qurify) that supplied the model its schema and business knowledge, auditing entity-merge accuracy and similarity thresholds to surface missing and duplicate nodes, and corrected errors in the reference answer key by reconciling output against source financial statements

Asan Medical Center (AMC) May 2025 - Aug 2025
AI Research Intern *Seoul, Republic of Korea*
- Built automated preprocessing pipelines for 3D CT imaging and clinical records, standardizing ingestion across modalities and cutting manual data preparation time for the research team
- Ran large-scale exploratory analysis in Python and Jupyter for the VLM3D Multi-Abnormality Classification Challenge, identifying the data patterns that informed feature selection and a strategy placing 7th among multi-institutional teams
- Co-authored a research paper on a multimodal deep survival model for hepatocellular carcinoma, applying SHAP and Grad-CAM to validate feature importance and communicate model reasoning to clinicians

Invention Corps of Berkeley; Berkeley Consulting Collaborative Jan 2025 - Feb 2026
User Research & Design Consultant *Berkeley, CA*
- **Mozilla Firefox:** structured a five-theme survey instrument with a 12-person team and analyzed responses into the feature requirements behind the React and Vite extension the team built and delivered, tracking 50+ concurrent tabs by memory and CPU usage
- **Braven:** cleaned and visualized survey data with an 8-person team, then re-segmented by class year when the aggregate profile proved too coarse, producing four cohort personas that drove the wireframes and 6 production-ready mobile screens delivered in Figma

PROJECTS

- **Multi-Agent Ad Generation System:** designed a versioned JSON spec as the contract between 8 pipeline stages, backed by a SQLite audience store (500 users, 12 segments), embedding-based retrieval, and deterministic validation gates
- **AI Tactical Assistant:** trained a PyTorch model on collected historical player and team data to predict match outcomes, then fed those predictions into a PuLP linear-programming optimizer that generated constrained optimal formations, used to win a 6v6 league
- **Activation Domain Prediction Pipeline:** developed a CNN-BiLSTM model classifying transcriptional activation domains from protein sequences with sliding-window feature engineering, SHAP interpretability, and unsupervised K-means clustering, reaching F1 = 0.95 across 10 distinct subclasses

LEADERSHIP & AWARDS

US Army, 8th Army 2ID HHBN Sep 2022 - Mar 2024
Non-Commissioned Officer *Republic of Korea*
- Led a 40+ soldier platoon as a bilingual Korean-English platoon leader to a 96% Army Combat Fitness Test pass rate